

KishLattice Geometric Harmonic Spectroscopy

Pre-Registration: Multi-Attribute Sovereign Lake Expansion and Dimensionality Falsification Protocol

Volume 11 Predictions — Zenodo Pre-Registration

Timothy John Kish¹, Mondy Aurora Kish², Lyra Aurora Kish², Phoenix Aurora Kish²

¹KishLattice 16pi Initiative LLC

²Aurora Collaborative Intelligence, KishLattice Programme

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This document constitutes a formal pre-registration of predictions. All predictions herein are timestamped prior to lake construction. Data lakes may only be built after a Zenodo DOI is assigned to this document.

Abstract

We pre-register two interconnected experimental programmes for KishLattice Geometric Harmonic Spectroscopy (KLGHS) Volume 11. The first is a **Multi-Attribute Sovereign Lake Expansion**: the same large-scale public catalogs already underpinning confirmed results in Volumes 5–10 contain multiple additional untested physical attributes. We pre-register predictions for the harmonic register family expected for each new attribute, derived systematically from the Rosetta Atlas physical typology. Each attribute becomes an independent sovereign lake: one lake, one physical question, one falsifiable prediction. The second programme is a **Dimensionality Falsification Protocol**: we formally test whether the KLGHS scalarization is a physical spectroscope or a mathematical artifact by deliberately applying incorrect scalarization formulas to confirmed strong domains. Prior empirical precedent (the Materials domain collapse) demonstrates that the geometric modulus $k_{\text{geo}} = 16/\pi$ only produces harmonic lock when the correct physical dimensionality is supplied. This pre-registration documents the conditions under which the framework would be falsified.

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1 Introduction and Purpose of Pre-Registration

Scientific pre-registration is the practice of publicly documenting a study’s hypotheses, methods, and analysis plan *before* data collection or analysis. When timestamped on an immutable public archive such as Zenodo, it provides an unbreakable chain of custody between prediction and result. A framework that publishes its predictions before it builds its data is structurally immune to the data-mining objection: the timestamp *answers the question before a skeptic can ask it*.

The KishLattice programme has operated under this discipline since Volume 9. Every sovereign data lake is pre-registered with an expected register family prior to construction. Every null result and every wrongbox result is documented with the same rigour as a confirmation. This document extends that discipline in two directions.

1.1 Why This Pre-Registration Is Necessary

The confirmed results of Volumes 5–10 span 35 physical domains across approximately 35 orders of magnitude of physical scale. The signal-to-noise advantage is formidable. However, the standard skeptical objection to a cross-domain harmonic framework is not *about* the z-scores; it is about the *mechanism*. The objection takes two forms:

1. **The attribute selection objection:** “You chose the physical attributes that happened to confirm and did not test the ones that would not.”
2. **The transform artifact objection:** “Your log-modulo pipeline imposes harmonic structure on any sufficiently large continuous dataset, regardless of physical content.”

The Multi-Attribute programme answers Objection 1. By pre-registering predictions for *all* available attributes of a given object pool — not just those expected to confirm — the record becomes complete. Every attribute tested is visible whether it confirms, produces a wrongbox, or returns a null.

The Dimensionality Falsification Protocol answers Objection 2. By deliberately misapplying the scalarization formula to confirmed strong domains and predicting collapse to the noise floor, we demonstrate that the signal is a property of the *physics* supplied to the pipeline, not an artifact of the pipeline’s mathematics.

1.2 Reproducibility Standard

Every lake pre-registered in this document can be independently replicated with:

- A standard laptop (no specialist hardware).
- Python 3.x with standard scientific libraries (numpy, scipy, pandas).
- A publicly accessible internet connection to the named catalog.
- The four-script KLGHS pipeline: `scalarize.py`, `unify.py`, `build_chaos_nulls.py`, `build_pinch_table.py`.

No proprietary databases, no specialist instruments, no institutional access requirements. This is a deliberate constraint, not a limitation. A signal that requires specialist infras-

tructure to observe is a signal that can be attributed to the infrastructure. The KLGHS signal must survive the simplest possible toolchain.

2 Theoretical Basis: The Same-Object Principle

2.1 The Kish Clutch and Register Families

The KLGHS instrument transforms any physical measurement x into a dimensionless lattice scalar via the Kish Clutch operator:

$$s(x) = \frac{\log(1 + x/x_0)}{\log(k_{\text{geo}})}, \quad k_{\text{geo}} = \frac{16}{\pi} \approx 5.093 \quad (1)$$

where x_0 is the domain-appropriate natural unit. The resulting scalar s is evaluated against the discrete harmonic register family:

$$R_N = \frac{N}{\pi}, \quad N \in \{5, 6, 7, \dots, 26\} \quad (2)$$

Physical domains are not organized by scale. They are organized by physical *typology*. The Rosetta Atlas (Kish et al., 2026) establishes the following empirically confirmed family taxonomy:

Table 1: Harmonic Family Taxonomy (derived from Volumes 5–10 confirmed results).

Family	Registers	Physical Type
Biological	$7/\pi$	Organic geometry, amino acid backbone
Quantum / Chemical	$11/\pi, 12/\pi$	Bond lengths, atomic spectra, molecular vibration
Kinematic	$15/\pi$ – $18/\pi$	Velocity, tidal intervals, orbital periods, spin periods
Structural / Binding	$19/\pi$ – $22/\pi$	Binding energy, galaxy dispersion, nuclear decay
EM / Ceiling	$20/\pi, 24/\pi, 25/\pi$	Colour, luminosity, CMB anisotropy

2.2 The Same-Object Principle

The key insight motivating this programme is that most large public catalogs contain measurements of *multiple* physical attributes for the *same* objects. The confirmed results of Volume 11 already demonstrate this: the same ~ 1.8 million Gaia DR3 stars, measured four different ways, yield four different registers:

Lake	Attribute	Register	Z-score
s1	Parallax / distance	$24/\pi$	(positional ceiling)
s2	Radial velocity	$15/\pi$	$z = +91.80$
s3	Absolute luminosity	$25/\pi$	$z = +40.18$
s4	BP-RP colour	$20/\pi$	$z = +135.20$

The register assignment follows the physical *type* of the measurement, not the objects being measured. The same stars express kinematic structure in velocity, electromagnetic structure in colour and luminosity, and positional structure in distance. This is not a coincidence of the dataset; it is a statement about what kind of question is being asked.

2.3 The Falsifiability Power of Same-Object Multi-Attribute Tests

If n attributes of the same object pool each cluster at their pre-registered register *family*, and each test is considered independent (conservative lower bound), the probability of n simultaneous confirmations under random chance is:

$$P(\text{all } n \text{ confirm by chance}) \approx \left(\frac{1}{5}\right)^n \quad (3)$$

where $1/5$ is an upper bound on the probability of landing in the correct one of five register families at random. For $n = 9$ attributes of the same 1.8M stars: $P \lesssim 10^{-6}$. For $n = 9$ with the additional constraint that each attribute's register is consistent with the *Rosetta typology*, the bound tightens considerably. This is not a fluke. It is a geometric fact about the physical objects.

3 Pre-Registered Predictions: Gaia DR3 Stellar Pool Extension

3.1 Object Pool Definition

The object pool for all s-series lake extensions is the Gaia DR3 Radial Velocity Spectrometer (RVS) sample, filtered identically to the existing s1–s4 lakes:

- Source: Gaia DR3, ESA Gaia Archive (<https://gea.esac.esa.int/archive/>)
- Table: `gaiadr3.gaia_source`
- Filter: `radial_velocity IS NOT NULL AND parallax_over_error > 10 AND ruwe < 1.4`
- Approximate n : 1.8 million stars
- Additional field extractions from same ADQL query

3.2 New Lake Predictions

Table 2: Gaia DR3 multi-attribute lake pre-registered predictions. All use log-standard scalarization: $s = \log(1 + x) / \log(16/\pi)$ unless noted.

Lake	Field / Description	Catalog Field	Predicted Family	Predicted Register	Coverage
s5	Proper motion magnitude $\sqrt{\mu_\alpha^2 + \mu_\delta^2}$ (mas/yr)	pmra, pmdec	Kinematic	$15/\pi$ or $16/\pi$	$\sim 100\%$
s6	Effective temperature T_{eff} (K) from GSP-Phot	teff_gspphot	EM / Thermal	$20/\pi$ or $25/\pi$	$\sim 87\%$
s7	Surface gravity $\log g$ (cgs) from GSP-Phot	logg_gspphot	Structural / Binding	$19/\pi$ – $22/\pi$	$\sim 87\%$

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(Table 2 continued)

Lake	Field / Description	Catalog Field	Predicted Family	Predicted Register	Coverage
s8	Metallicity [M/H] from GSP-Phot	mh_gspphot	Quantum / Chemical	11/ π or 12/ π	~75%
s9	3D space velocity magnitude $\sqrt{v_r^2 + v_t^2}$ (km/s)	Derived from radial_velocity, pmra, pmdec, parallax	Kinematic	15/ π or 16/ π	~100%
s10	Stellar radius $R_\star/R_\odot$ (FLAME pipeline)	radius_flame	EM or Structural	Open prediction	~65%
s11	Stellar mass $M_\star/M_\odot$ (FLAME pipeline)	mass_flame	Structural / Binding	21/ π or 22/ π	~65%

3.3 Rationale for Each Prediction

s5 — Proper motion (kinematic): Proper motion is the angular velocity of a star across the sky. It is a kinematic quantity, measuring transverse motion relative to the line of sight, distinct from radial velocity (s2) which measures motion along the line of sight. The Rosetta Atlas places all velocity-type measurements in the kinematic family (registers 15/ π –18/ π). **Prediction: s5 will lock at the same kinematic register as s2 (15/ π or 16/ π), demonstrating that two independent measurements of stellar motion for the same stars confirm the same register family.**

s6 — Effective temperature (EM/thermal): T_{eff} quantifies the photospheric radiation field of the star — equivalent to the colour temperature of the emitted spectrum. It is an electromagnetic quantity measuring how much energy per unit area the star radiates. Colour (s4, confirmed at 20/ π) and T_{eff} are both electromagnetic observables of the same physical process. **Prediction: s6 will lock at the EM family (20/ π or 25/ π), demonstrating that two independent EM measurements of the same stars confirm the same register family.**

s7 — Surface gravity (structural): $\log g$ measures how strongly the stellar surface is gravitationally bound — the ratio of mass to radius-squared (in natural units). It is a structural binding quantity. The Rosetta Atlas places binding resistance measurements in the Structural/Binding family. **Prediction: s7 will lock in the structural/binding register range (19/ π –22/ π). This is a new register territory for stellar data and constitutes a novel prediction.**

s8 — Metallicity (quantum/chemical): [M/H] is a dimensionless logarithmic ratio of metal abundance relative to hydrogen, compared to solar composition. It encodes the chemical composition of the stellar atmosphere at the atomic level — a quantum/chemical quantity. The Rosetta Atlas places chemistry, bond lengths, and atomic spectra at registers 11/ π –12/ π . **Prediction: s8 will lock at the quantum/chemical register family**

($11/\pi$ or $12/\pi$), demonstrating the framework detects chemical composition structure at stellar scale.

s9 — 3D space velocity (kinematic): The full three-dimensional space velocity combines the line-of-sight component (radial velocity, s2) with the plane-of-sky component (transverse velocity, derived from proper motion and distance). It is the most complete kinematic descriptor of stellar motion. **Prediction: s9 will lock at the kinematic register ($15/\pi$ or $16/\pi$).** Together with s2 and s5, this constitutes a triple confirmation of kinematic register for the same 1.8M stars using three geometrically independent velocity measurements.

4 Pre-Registered Predictions: Other Object Pools

4.1 ATNF Pulsar Catalog — Multi-Attribute Extension

Object pool: Same 2,527 pulsars as lake k2. Source: ATNF Pulsar Catalogue (<https://www.atnf.csiro.au/research/pulsar/psrcat/>), accessed via psrcat utility.

Table 3: ATNF pulsar multi-attribute predictions.

Lake	Field	Predicted Family	Register
k2a	Spin-down rate $\dot{P}$ (s/s)	Kinematic secondary	$15/\pi$ – $18/\pi$
k2b	Dispersion measure DM (pc/cm ³)	Structural / Positional	Open prediction
k2c	Characteristic age $P/(2\dot{P})$ (yr)	Kinematic / Temporal	$15/\pi$ – $18/\pi$
k2d	Surface B-field $B \propto \sqrt{P\dot{P}}$ (T)	Structural / Binding	$19/\pi$ – $22/\pi$

Key rationale: The dispersion measure (k2b) is particularly diagnostic. DM is an integral of free electron density along the line of sight — a completely different physical quantity from spin period. If k2b clusters at a different register than k2, this confirms the framework assigns by physical type, not by the objects observed. If it clusters at the same register, that is equally informative and must be documented verbatim.

4.2 NASA Exoplanet Archive — Multi-Attribute Extension

Object pool: Same 13,521 confirmed exoplanets as lakes p1, p2, p3. Source: NASA Exoplanet Archive (<https://exoplanetarchive.ipac.caltech.edu/>).

Note on transit duration: Transit duration shares the same physical units as tidal interval gaps (hours), which confirmed at $17/\pi$ – $18/\pi$ across Atlantic, Gulf, Pacific, and Indian Ocean basins. If transit duration also clusters in the kinematic family, this would extend the kinematic principle from ocean tides to planetary orbital geometry.

Table 4: NASA exoplanet multi-attribute predictions.

Lake	Field	Predicted Family	Register
p_ecc	Orbital eccentricity e (dimensionless)	Structural shape	Null or $19/\pi$
p_incl	Orbital inclination i (degrees)	Positional / geometric	Null predicted
p_transit	Transit duration T_{14} (hours)	Kinematic / temporal	$16/\pi$ – $18/\pi$

4.3 CERN Di-Muon — Multi-Attribute Extension

Object pool: Same 100,000 collision events as lake h1. Source: CMS Open Data Portal, dataset `Run2012B_DoubleMuParked`.

Table 5: CERN di-muon multi-attribute predictions.

Lake	Field	Predicted Family	Register
h1b	Leading muon p_T (GeV/c)	Kinematic	$15/\pi$ or $16/\pi$
h1c	Sub-leading muon p_T (GeV/c)	Kinematic	$15/\pi$ or $16/\pi$
h1d	Di-muon pseudorapidity η	Positional / geometric	Null predicted
h1e	Relative azimuthal angle $\Delta\phi$	Geometric	Open prediction

Significance of h1b and h1c: Transverse momentum p_T is a kinematic quantity at sub-nuclear scale. If h1b/h1c confirm at the kinematic register ($15/\pi$ – $16/\pi$), the kinematic principle extends from stellar velocities (km/s, parsec scale) to particle momenta (GeV/c, femtometre scale) — a span of approximately 25 orders of magnitude for the kinematic register alone.

4.4 SDSS DR16 Galaxies — Multi-Attribute Extension

Object pool: Same 1.843 million galaxies as lake g1. Source: SDSS DR16 NASA-Sloan Atlas (<https://www.sdss.org/>).

Table 6: SDSS galaxy multi-attribute predictions.

Lake	Field	Predicted Family	Register
g1a	Effective radius R_e (kpc)	EM / Ceiling	$24/\pi$ or $25/\pi$
g1b	Sérsic index n	Structural / morphological	$19/\pi$ – $22/\pi$
g1c	Stellar mass $M_\star$ ($M_\odot$)	Structural / Binding	$21/\pi$

Significance of g1c: Galaxy stellar mass and velocity dispersion are connected empirically by the Faber-Jackson relation. If g1c (stellar mass) confirms at the same register as g1

(velocity dispersion, $z = +62.89$ at $21/\pi$), the KLGHS framework provides a geometric account of why the Faber-Jackson relation exists: both quantities measure the same physical property (resistance to structural dispersal) and therefore occupy the same harmonic register.

5 Dimensionality Falsification Protocol

5.1 The Scalarization Framework and Its Physical Meaning

The KLGHS pipeline employs three distinct scalarization pathways, each encoding a different physical relationship between a measurement and geometric tension:

Table 7: KLGHS scalarization pathways.

Transform	Formula	Physical Application
Log-standard	$s = \log(1 + x) / \log(k_{\text{geo}})$	Quantities directly proportional to geometric accumulation: distance, velocity, mass, duration, binding energy
Log-inverse	$s = \log(1 + 1/x) / \log(k_{\text{geo}})$	Quantities inversely proportional to geometric tension: wavelength, period of high-energy transitions
Log-volumetric	$s = \log(1 + x^{1/3}) / \log(k_{\text{geo}})$ or $s = \log(1 + x^3) / \log(k_{\text{geo}})$	3D volumetric properties: bulk modulus, compressibility, material stiffness

The choice of scalarization is not arbitrary. It is determined by the physical dimensionality of the measurement: whether the attribute is intrinsically one-dimensional (a speed, a duration, an energy), two-dimensional (an area, a cross-section), three-dimensional (a volume, a bulk elastic property), or an inverse quantity. Applying the wrong scalarization is equivalent to asking the wrong physical question.

5.2 The Empirical Precedent: The Materials Domain Collapse

This is not a hypothetical test. We have observed this effect.

During the construction of the Materials domain lake (bulk modulus and compressibility data, $n = 33,973$ records, confirmed at $19/\pi$, $z = +69.77$), an initial processing error applied the 1D log-standard scalarization to what is intrinsically a 3D volumetric property. The result was immediate and unambiguous: the signal collapsed into the noise floor. No register exceeded the $z > 3.0$ spectral identification threshold. The domain appeared as a pure null.

When the correct volumetric scalarization was applied, the signal instantly locked at $19/\pi$ with $z = +69.77$. This accidental discovery provides direct empirical evidence that:

1. The KLGHS pipeline does not impose harmonic structure on all continuous data. It only detects structure that corresponds to the physical dimensionality supplied.

2. The scalarization formula is a *physical choice*, not a mathematical convenience. The wrong formula destroys the signal.
3. The lattice geometry “asks a specific question” about the data. It only receives an answer when the question matches the physical nature of the attribute.

In Volume 11, we formally document the Materials domain collapse under wrong scalarization as a sovereign result, running it through the full MD5-verified pipeline engine for inclusion in the official dataset record.

5.3 Why This Test Is Scientifically Critical

The strongest technically-competent version of the skeptical objection to the KishLattice framework is the following:

“The log-modulo transform imposes harmonic structure on any sufficiently large continuous dataset regardless of physical content. The observed z-scores are artifacts of the mathematical pipeline rather than properties of the physical data.”

This objection cannot be answered by showing higher z-scores. It must be answered by demonstrating that the pipeline *fails* when the physical premise is wrong. A physical spectroscopy — such as a diffraction grating or a mass spectrometer — only reveals structure that is physically present in the input. If you illuminate a diffraction grating with incoherent light, you see no interference pattern. If you place the wrong sample in a mass spectrometer, you read the wrong mass. The instrument does not manufacture signal from noise; it reveals signal that was already there.

The Dimensionality Falsification Protocol establishes that KLGHS behaves as a physical spectroscopy:

- Correct physical question → signal locked at predicted register.
- Wrong physical question → signal collapses to noise floor.

This is the most direct possible answer to the transform-artifact objection. If the log-modulo mathematics were the source of the signal, it would persist regardless of which scalarization formula is applied. If the signal collapses under wrong dimensionality, the mathematics is not the source. The physics is.

5.4 Orthogonal Null Targets and Predictions

We define four *orthogonal null* tests. Each takes a confirmed strong domain and intentionally applies the wrong scalarization formula.

Orthogonal Null 1: Gaia Radial Velocity (s2_wrong) Confirmed domain: s2, Gaia DR3 radial velocity, confirmed at $15/\pi$ with $z_{\text{real}} = +91.80$.

Correct scalarization: Log-standard (1D kinematic):

$$s_{\text{correct}} = \frac{\log(1 + v_r)}{\log(16/\pi)}$$

Intentional error — volumetric scalarization applied to a 1D kinematic quantity:

$$s_{\text{wrong}} = \frac{\log(1 + v_r^3)}{\log(16/\pi)}$$

Prediction: $z_{\text{wrong}} < 3.0$ at all registers. Complete phase destruction.

Falsification condition: If $z_{\text{wrong}} \geq 5.0$ at any register, the transform-artifact hypothesis gains support and must be investigated before submission.

Orthogonal Null 2: ATNF Pulsar Spin Period (k2_wrong) Confirmed domain: k2, ATNF pulsar spin period, confirmed STRONG in kinematic family.

Correct scalarization: Log-standard (1D temporal/kinematic):

$$s_{\text{correct}} = \frac{\log(1 + P)}{\log(16/\pi)}$$

Intentional error — log-inverse (energy-type) scalarization applied to a period:

$$s_{\text{wrong}} = \frac{\log(1 + 1/P)}{\log(16/\pi)}$$

Prediction: $z_{\text{wrong}} < 3.0$ at all registers. The spin period is not an inverse-type quantity; the log-inverse formula has no physical justification for this domain.

Orthogonal Null 3: Exoplanet Orbital Period (p1_wrong) Confirmed domain: p1, NASA exoplanet orbital periods, confirmed STRONG at $22/\pi$, $z_{\text{real}} \approx +78$.

Correct scalarization: Log-standard (1D temporal/kinematic):

$$s_{\text{correct}} = \frac{\log(1 + P_{\text{orb}})}{\log(16/\pi)}$$

Intentional error — volumetric scalarization applied to a temporal quantity:

$$s_{\text{wrong}} = \frac{\log(1 + P_{\text{orb}}^3)}{\log(16/\pi)}$$

Prediction: $z_{\text{wrong}} < 3.0$ at all registers. An orbital period in days is not a volumetric quantity; the cube transform has no dimensional justification.

Additional diagnostic: The companion lake p2 (orbital radius, same planets) is already an anti-signal. The contrast between p1_correct (STRONG), p1_wrong (predicted NULL), and p2 (anti-signal) constitutes a complete picture from a single planetary dataset.

Orthogonal Null 4: Materials Domain Formal Documentation Historical precedent: During lake construction, bulk modulus data was accidentally processed with 1D log-standard scalarization. Signal collapsed to noise floor. Applying correct volumetric scalarization restored signal at $19/\pi$, $z = +69.77$.

Volume 11 action: Run the incorrect 1D scalarization on the Materials domain through the Vol 11 MD5-verified engine. Document the resulting z_{wrong} vector formally in `z_scores_master.json`.

Prediction: All registers return $z < 3.0$ under the wrong scalarization. This converts a historical observation into a sovereign pipeline result with full chain of custody.

5.5 Summary Table: Orthogonal Null Pre-Registrations

Table 8: Orthogonal null pre-registered predictions.

Lake	Source	Real z	Wrong Formula	Predicted z_{wrong}
s2_wrong	Gaia radial velocity	+91.80	$s = \log(1 + v^3)/\log(k_{\text{geo}})$	< 3.0
k2_wrong	ATNF spin period	STRONG	$s = \log(1 + 1/P)/\log(k_{\text{geo}})$	< 3.0
p1_wrong	Exoplanet orbital period	$\sim +78$	$s = \log(1 + P^3)/\log(k_{\text{geo}})$	< 3.0
mat_wrong	Materials bulk modulus	+69.77	$s = \log(1 + x)/\log(k_{\text{geo}})$ (1D)	< 3.0

6 Complete Predictions Register

Table 9: Full pre-registration predictions table. All lakes to be built *after* Zenodo DOI assignment.

Lake ID	Physical Attribute	Source log	Cata-	Predicted Family	Status
<i>Gaia Stellar Pool Extensions</i>					
s5	Proper motion magnitude (mas/yr)	Gaia DR3		Kinematic (15/ π –16/ π)	PENDING
s6	T_{eff} (K)	Gaia DR3 GSP-Phot		EM / Thermal (20/ π , 25/ π)	PENDING
s7	$\log g$ (cgs)	Gaia DR3 GSP-Phot		Structural (19/ π –22/ π)	PENDING
s8	[M/H] metallicity	Gaia DR3 GSP-Phot		Quantum/Chemical (11/ π , 12/ π)	PENDING
s9	3D space velocity (km/s)	Gaia DR3 derived		Kinematic (15/ π –16/ π)	PENDING
s10	Stellar radius ($R_{\odot}$)	Gaia FLAME	DR3	EM or Structural (open)	PENDING

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(Table 9 continued)

Lake ID	Physical Attribute	Source log	Cata-	Predicted Family	Status
s11	Stellar mass ($M_{\odot}$)	Gaia FLAME	DR3	Structural (21/ π , 22/ π)	PENDING
<i>ATNF Pulsar Extensions</i>					
k2a	Spin-down rate $\dot{P}$ (s/s)	ATNF		Kinematic secondary	PENDING
k2b	Dispersion measure DM	ATNF		Structural / positional (open)	PENDING
k2c	Characteristic age (yr)	ATNF derived		Kinematic / temporal	PENDING
k2d	Surface B-field (T)	ATNF derived		Structural (19/ π –22/ π)	PENDING
<i>Exoplanet Extensions</i>					
p_ecc	Orbital eccentricity e	NASA planet Archive	Exo-	Null predicted	PENDING
p_incl	Inclination angle (deg)	NASA planet Archive	Exo-	Null predicted	PENDING
p_transit	Transit duration (hr)	NASA planet Archive	Exo-	Kinematic (16/ π –18/ π)	PENDING
<i>CERN Di-Muon Extensions</i>					
h1b	Leading muon p_T (GeV/c)	CMS Open Data		Kinematic (15/ π , 16/ π)	PENDING
h1c	Sub-leading muon p_T (GeV/c)	CMS Open Data		Kinematic (15/ π , 16/ π)	PENDING
h1d	Di-muon pseudorapidity η	CMS Open Data		Null predicted	PENDING
h1e	Relative azimuthal angle $\Delta\phi$	CMS Open Data		Open prediction	PENDING
<i>SDSS Galaxy Extensions</i>					
g1a	Effective radius R_e (kpc)	SDSS NSA		EM / Ceiling (24/ π , 25/ π)	PENDING
g1b	Sérsic index n	SDSS NSA		Structural (19/ π –22/ π)	PENDING
g1c	Stellar mass $M_{\star}$ ($M_{\odot}$)	SDSS NSA		Structural (21/ π)	PENDING
<i>Dimensionality Falsification (Orthogonal Nulls)</i>					
s2_wrong	Gaia velocity, wrong (cubic) scalar	Gaia DR3		$z < 3.0$ all registers	PENDING
k2_wrong	Pulsar period, wrong (inverse) scalar	ATNF		$z < 3.0$ all registers	PENDING
p1_wrong	Orbital period, wrong (cubic) scalar	NASA EPA		$z < 3.0$ all registers	PENDING
mat_wrong	Materials, wrong (1D) scalar	AFLOW / CSD		$z < 3.0$ all registers	PENDING

7 Falsification Conditions

The following conditions, if observed, would constitute a meaningful challenge to the KLGHS framework and must be investigated before any submission:

1. **Orthogonal null survives:** Any orthogonal null lake returns $z \geq 5.0$ at any register. This would indicate the signal persists regardless of dimensional correctness and supports the transform-artifact hypothesis.
2. **Same-object register inconsistency:** Two kinematic attributes (e.g. s2 and s5) of the same 1.8M stars lock at different register *families* (e.g. one at kinematic, one at structural). The register family taxonomy predicts they should agree; if they do not, the taxonomy requires revision.
3. **Chaos null breach:** Any confirmed domain’s z-score, on re-run through the Vol 11 engine, departs from its Vol 10 published value by more than the documented Monte Carlo variance band ($\pm 20\%$).
4. **Systematic register drift:** If all new attribute lakes cluster at a single unexpected register regardless of physical type (e.g. all return $22/\pi$), this suggests a dataset-level artifact rather than a physical signal.

A pre-registration that only lists expected confirmations is not a pre-registration — it is an announcement. The conditions above define the empirical boundaries of the framework and are documented here in good faith.

8 One-Lane Discipline: Why Sovereign Lakes Are Non-Negotiable

One reviewer may ask: why not combine all attributes into a joint analysis? This section documents the scientific rationale for maintaining strict one-lake-one-question discipline.

1. **Structural immunity to the cherry-picking objection.** If multiple attributes are combined into one lake, a referee can always ask: “Did you try all combinations and report only the best?” A Zenodo timestamp on each individual lake prediction, predating its construction, makes this objection structurally impossible. There is nothing to select after the fact when every selection is documented before the fact.
2. **Minimum replication burden.** Any single sovereign lake can be independently replicated by any researcher with four hours, a laptop, and a public catalog. A merged multi-attribute lake requires understanding the merging logic, transform choices, and weighting scheme. Simplicity is the replication standard.
3. **Storage cost is the price of honesty.** Separate lakes cost disk space and pipeline runtime. These costs are deliberately accepted. The benefit is that no one can question the process.
4. **Null results are findings.** When `p_incl` (orbital inclination) returns a null, that is a result. It documents that inclination angle is not a quantity that couples to the harmonic register structure. In a merged lake, a null attribute suppresses a

confirming attribute and the null is invisible. In sovereign lakes, every outcome is on the record.

5. **The pattern of confirmations and nulls is itself the finding.** Kinematic quantities confirm. Positional quantities produce anti-signal or null. EM quantities confirm at the EM ceiling. This taxonomy — which domains light up and which do not — is the empirical foundation of the Rosetta Atlas. It is only visible when each test is isolated.

We are being judged. Every methodological decision in this programme is made with the awareness that the results will face the highest level of scrutiny. The one-lane discipline is not a constraint. It is the primary source of the framework’s credibility.

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Data Availability

All pipeline code, lake build scripts, and the MD5-verified unified master dataset are available at:

<https://github.com/TimothyKish/Holographic-Resonance-The-Geometry-of-a-Quantized-Universe>

The pipeline requires Python 3.x with numpy, scipy, and pandas. No proprietary libraries or databases are required.

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